

Module 05 - Plans, Trips & Itinerary

Main collaboration object for activities and travel

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

A plan represents a shared activity: trip, movie, dinner, event, or custom plan. It contains members, timing, location, optional itinerary items, and links to bookings, expenses, polls, and activity events.

Plan document

```
groups/{groupId}/plans/{planId}
{
  title,
  type: "trip" | "movie" | "dinner" | "event" | "other",
  description,
  startAt,
  endAt,
  location,
  coverImageUrl,
  status: "draft" | "active" | "completed" | "cancelled",
  createdBy,
  createdAt,
  updatedAt
}
```

Plan membership

```
groups/{groupId}/plans/{planId}/members/{uid}
{ uid, joinedAt, role: "organizer" | "participant", displayNameSnapshot }
```

Itinerary item

```
../plans/{planId}/itinerary/{itemId}
{
  title,
  type,
  startAt,
  endAt,
  location,
  notes,
  order,
  createdBy
}
```

Core flows

- Create plan
- Edit plan
- Add/remove participants
- Add/edit/delete itinerary item
- Reorder itinerary
- Mark plan completed/cancelled

Plan page tabs

Acceptance criteria

- Plan creation validates required title/type.
- Only valid group members can be attached to plan.
- Organizer/admin permissions are enforced.
- Itinerary order persists after refresh.
- Completed/cancelled plans remain readable for history.